EXPERIMENT 15

Vehicle Service Request Management System

Future Scope, ApplicationWorkflow,QualityControlsandCareerPathways in Low-Code Development

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Experiment	15 — Final Experiment Report
Project Topic	Vehicle Service Request Management System

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A practical study of how a low-code application can capture service requests, support workshop coordination, maintain traceable status updates, and evolve through secure integrations and automation.

Submitted as a practical report

The interface illustrations in this document are representative mock-ups created to explain the proposed system workflow. They are not screenshots of a deployed production application.

1. Aim and Scope

Toexamine howaVehicle Service Request Management System can be planned and improved using a low-code development approach. The experiment connects the core application workflow with future enhancements, quality practices, automation opportunities, and transferable career skills.

Objectives

- Identify the main modules required to register and track vehicle-service requests.
- Describe the flow from customer submission through staff review, technician assignment, and completion.
- Explore practical improvements such as validation, notifications, reporting, and integration with other services.
- Recognise security, data-quality, and governance controls needed for a reliable application.
- Map the project experience to future learning in low-code development, integration, automation, and business analysis.

2. Procedure

Step	Activity	Description
1	Review the scenario	Identify the customer, vehicle, service type, preferred date, and urgency details required for a request.
2	Plan the data capture	Define required fields and validation rules to reduce incomplete or inconsistent submissions.
3	Design the workflow	Map request submission, staff review, assignment, service progress, and closure.
4	Prepare application views	Create a dashboard, request form, status board, and report view for different user needs.
5	Review quality controls	Consider access permissions, duplicate prevention, audit history, and safe handling of customer information.
6	Plan future improvements	Prioritise notifications, appointment scheduling, analytics, and secure integrations.

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3. System Workflow and Main Interface

The proposed system centralises service requestsinastructuredrecord. A customer submits vehicle and service details; staff validate the request and assign a technician; progress is updated through defined statuses; and reports help the service centre review workload and completed jobs.

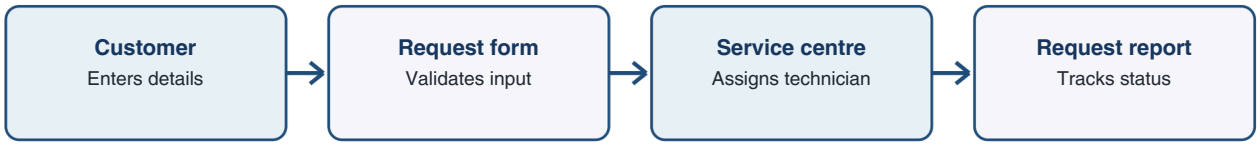


Figure 1. Proposed end-to-end workflow for the Vehicle Service Request Management System.



Figure 2. Illustrative admin dashboard showing request totals, current status, and recent service records.

Interpretation

The dashboard gives staff a quick operational summary. Status totals support workload review, while the recent-request list helps locate records for follow-up. In a real implementation, the displayed counts and sample requests would be populated from the application database and should reflect live data only after integration and testing.

4. Request Form and Data Requirements

Aclear requestformreduces repeatedcallsandmakesit easier for the service centre to understand the customer's needs before confirming an appointment. Required fields should be limited to information necessary to process the request.



Figure 3. Representative vehicle-service request form with example values; not a live application screenshot.

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Recommended fields and validation

Field	Purpose	Validation / handling
Customer name	Identify the requester	Required; trim unnecessary spaces.
Vehicle registration	Link the request to a vehicle	Required; apply a consistent format.
Vehicle model	Help staff identify the vehicle	Use a controlled list where practical.
Service required	Describe the requested work	Select from service options; allow notes if needed.
Preferred date	Support appointment planning	Check date format and available booking rules.
Priority / urgency	Help staff triage requests	Use defined values and approved priority rules.
Request status	Track progress	Restrict changes to authorised roles.

5. Future Enhancements and Quality Controls

Thefollowing features are recommendedasfuturework.Theyshouldbetreated as planned enhancements until implemented and tested in the application.

Future enhancement plan

Enhancement	Expected value	Key consideration
Appointment-slot selection	Helps customers request a suitable time and reduces manual coordination.	Availability must be kept current.
Email / SMS notifications	Keeps customers informed about submission, assignment, and completion.	Use consent-aware messaging and verified contact details.
Service-cost estimate and invoice	Improves transparency around expected costs and completed work.	Estimates should be labelled and confirmed by staff.
Technician workload view	Helps supervisors distribute work fairly.	Assignment rules should reflect skills and capacity.
Customer feedback	Captures service experience and recurring issues.	Protect feedback records and review them consistently.
Analytics dashboard	Highlights request volume, delays, and service trends.	Use accurate timestamps and clearly defined metrics.

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Security and reliability checklist

- Authentication and role-based permissions for customers, technicians, and administrators.
- Input validation and duplicate-submission checks before creating a request.
- Audit history for important status changes and technician assignments.
- Secure handling of customer contact details and vehicle information.
- Error handling for failed notifications or external service connections.
- Regular testing of form rules, reports, and access permissions before release.

6. Skills and Career Pathways

Developing this system bringstogether business-process understanding and technical implementation. These skills can support progression into low-code development, business analysis, application support, integration development, and automation roles.

Skill area	Application in this project	Possible next step
Requirements analysis	Identify customer and service-centre needs, define request statuses.	Practise user stories, acceptance criteria, and process maps.
Low-code development	Configure forms, reports, views, and workflow rules.	Build a complete prototype with sample records.
Data management	Structure request, vehicle, customer, and technician records.	Learn relational modelling and SQL fundamentals.
Integration	Plan connections for email, SMS, calendars, or invoicing.	Practise REST APIs, JSON, and authentication.
Testing and governance	Check validation, permissions, and edge cases.	Create test cases and maintain change documentation.
Reporting and analytics	Review request volumes, delays, and completion rates.	Explore spreadsheets or a BI dashboard.

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7. Result

Theexperiment produced a structured design for a Vehicle Service Request Management System, including the service-request workflow, representative dashboard and form views, recommended data fields, future enhancements, quality controls, and a skill-development map. The report describes a design-level proposal; it does not claim that a production system has been deployed or benchmarked.

8. Conclusion

Awell-organised vehicle-service application can improve traceability by keeping customer, vehicle, service, assignment, and status information together. Low-code tools can accelerate the creation of forms and reports, while careful validation, permissions, testing, and governance remain essential. Future work should be prioritised around customer communication, appointment coordination, operational reporting, and secure integrations.

9. References

1. Oracle Academy, Application Development Foundations learning materials.
2. Zoho Creator help and product documentation, for low-code forms, reports, workflows, and application administration.
3. Oracle APEX and general database documentation, for broader application-development and data-management concepts.
4. Course notes and practical instructions provided for the experiment.

Report note: Interface images are illustrative mock-ups created specifically for the Vehicle Service Request Management System. Sample names, registration numbers, dates, and counts are fictional and are included only to explain the proposed layout.